

Wearable Posture Detection and Warning System

INF340 – Microprocessors – Term Project Proposal

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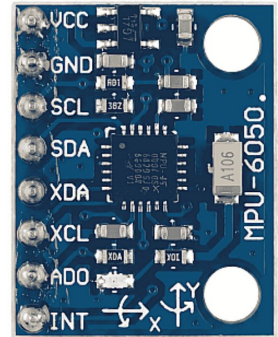
Ahmet Faruk Yetkin

Project Overview

We are building a wearable system that tracks the user's posture in real time and warns them when they slouch. We place two IMUs — one on the hip and one on the upper back — and the angle between them tells us how curved the spine is. The MSP432 fuses the sensor data, classifies the posture against a pre-calibrated “neutral” position, fires a buzzer when needed, and streams data over BLE to a host computer. On the PC side, the user's posture is shown as a stick figure and a posture profile is generated at the end of each session.

Hardware Components

- **MSP432P401R LaunchPad** — main microcontroller; sensor fusion and classification run here.
- **Piezo buzzer** — for the warning signal, connected directly to a GPIO pin.
- **2× MPU-6050 IMU modules (SY-104)** — accelerometer + gyroscope, sharing the same I²C bus.
- **HM-10 BLE module** — UART-to-BLE bridge for wireless data transmission.
- **Mounting:** ribbon cable between the sensors, elastic straps, a belt pouch to carry the LaunchPad and battery.



Deliverables

1. Wearable hardware with two IMUs and the MSP432.
2. MSP432 firmware in C handling sensor fusion, calibration, threshold-based posture classification, warning logic, and BLE transmission.
3. Python application providing real-time stick-figure visualization, session logging, and posture profile generation.

Division of Labour

Furkan Arda Karaman — on-device work: sensor fusion, classification logic, warning state machine, and PC-side posture profile analytics.

Ahmet Faruk Yetkin — hardware integration: IMU connections and drivers, BLE module integration, data transmission protocol, and PC-side data reception.

Ahmet Cemil Yalçındağ — PC application: real-time visualization, session logging, UI, and physical assembly of the wearable.

Possible Future Improvements

The classifier in this proposal is rule-based. Later on we could replace it with a small neural network to distinguish between postures that look similar but have different intent — like chronic slouching versus leaning forward to read something. We mention this as a direction we could build on later, not as part of what we will deliver.